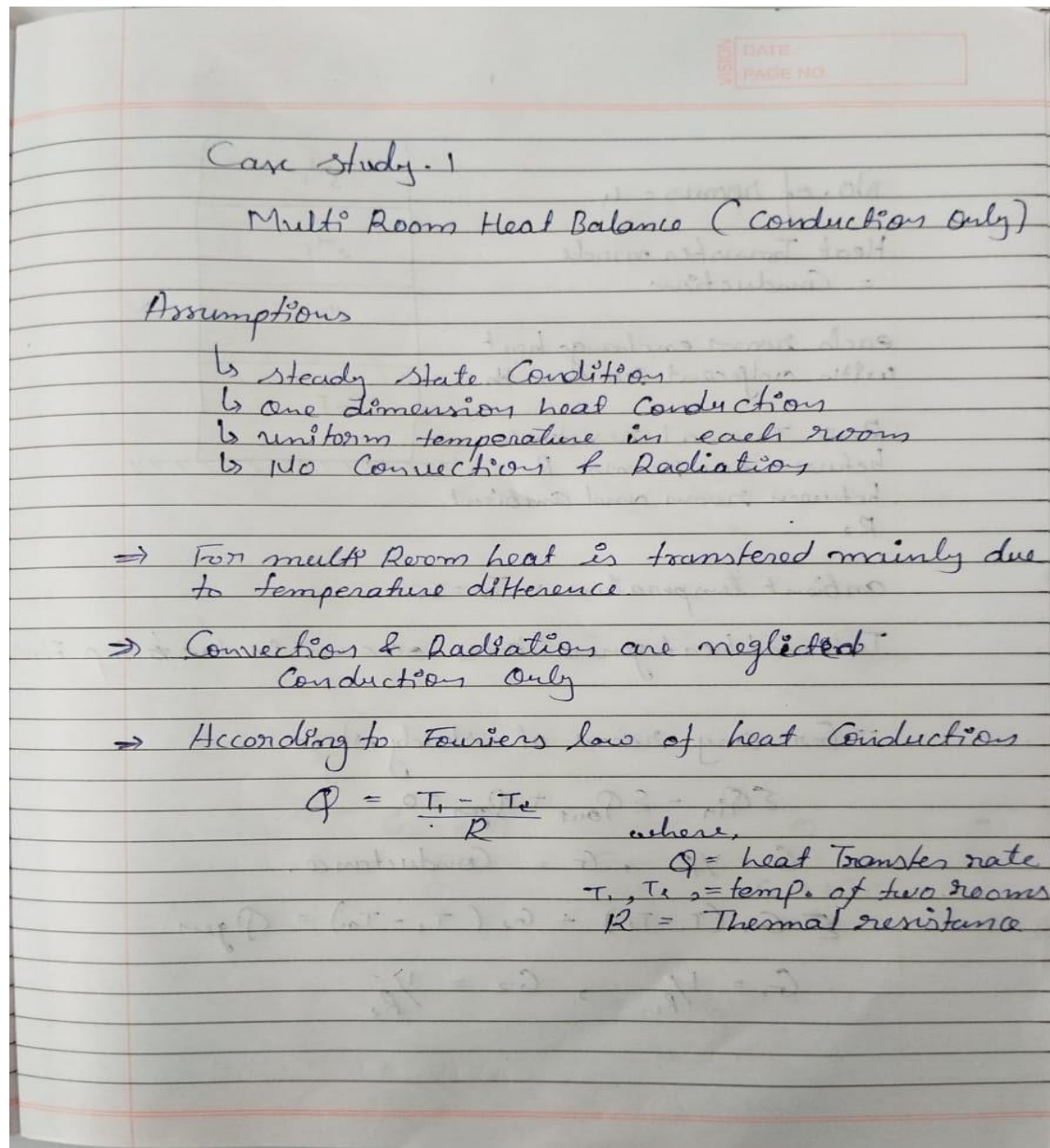


CASE STUDY 1: Multi Room Heat Balance (Conduction only)

INTRODUCTION

The multi-room heat balance method is used to analyze heat transfer between interconnected rooms and the surroundings. In this study, only conduction heat transfer through walls and partitions is considered. Each room is treated as a control volume, and energy balance equations are applied. The approach helps determine steady-state room temperatures based on thermal resistances.

Mathematical Expression

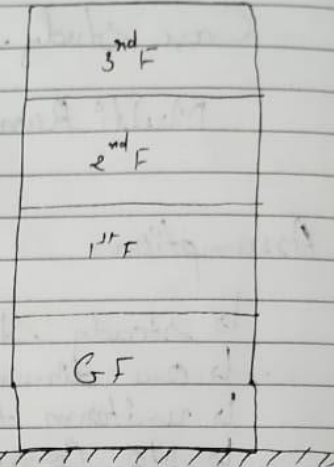


No. of rooms = 4

Heat Transfer mode
= Conduction

each room exchange heat
with adjacent rooms th

Req: Thermal resistance
between two rooms R_1
between room and ambient
 R_2



ambient temperature = $T_a = 35^\circ\text{C}$

Internal heat generation occurs in the top floor

$\therefore$ For any room at steady state

$$\sum \dot{Q}_{in} - \sum \dot{Q}_{out} + \dot{Q}_{gen} = 0$$

$$G = 1/R, \quad G = \text{Conductance.}$$

$$\sum G_1 (T_1 - T_2) + G_2 (T_1 - T_a) = \dot{Q}_{gen}$$

$$G_1 = 1/R_1, \quad G_2 = 1/R_2$$

⇒ Ground Floor

$$G_1 (T_0 - T_1) + G_2 (T_0 - T_a) = 0$$

$$\therefore (G_1 + G_2) T_0 - G_1 T_1 = G_2 T_a$$

⇒ Middle room (1 & 2)

$$G_1 (T_i - T_{i-1}) + G_1 (T_i - T_{i+1}) + G_2 (T_i - T_a) = 0$$

$$\therefore (G_2 + 2G_1) T_i - G_1 T_{i-1} - G_1 T_{i+1} = G_2 T_a$$

⇒ Top Floor

$$\therefore (G_1 + G_2) T_3 - G_1 T_2 = G_2 T_a + Q_{gen}$$

Matrix $[A][T] = [B]$

$$A = \begin{bmatrix} G_2 + G_1 & -G_1 & 0 & 0 \\ -G_1 & G_2 + 2G_1 & -G_1 & 0 \\ 0 & -G_1 & G_2 + 2G_1 & -G_1 \\ 0 & 0 & -G_1 & G_2 + G_1 \end{bmatrix}$$

$$T = \begin{bmatrix} T_0 \\ T_1 \\ T_2 \\ T_3 \end{bmatrix}$$

$$B = \begin{bmatrix} (G_1 + G_2) T_a \\ G_2 T_a \\ G_2 T_a \\ (G_1 + G_2) T_a + Q_{gen} \end{bmatrix}$$

CODE

```
clc;
clear;

R1 = 0.1;
R2 = 0.48;

Tamb = 35;

Qgen = 50;

G1 = 1/R1;
G2 = 1/R2;

A = [ (G2+G1)      -G1      0      0;
      -G1      (G2+G1+G1)  -G1      0;
      0      -G1      (G2+G1+G1)  -G1;
      0      0      -G1      (G2+G1) ];

B = [ Tamb*(G2);
      Tamb*(G2);
      Tamb*(G2);
      Tamb*(G2) + Qgen ]; T

T= inv(A)*B;

fprintf('Temperature at Ground Floor (T0) = %.2f °C\n', T(1));
fprintf('Temperature at 1st Floor (T1) = %.2f °C\n', T(2));
fprintf('Temperature at 2nd Floor (T2) = %.2f °C\n', T(3));
fprintf('Temperature at 3rd Floor (T3) = %.2f °C\n', T(4));
floors = 0:3;

figure;
plot(floors, T, '-o', 'LineWidth', 1.5, 'MarkerSize', 8);
grid on;
xlabel('Floor (0 = Ground)');
ylabel('Temperature (°C)');
title('Temperature vs Floor');
xlim([0 3]);
xticks(floors);

for k = 1:numel(T)
    text(floors(k), T(k), sprintf(' %.2f°C', T(k)),
        'VerticalAlignment','bottom');
end
```

